

AI Study Companion — AI Tools Documentation

Part A: PRD Requirements | Part B: Recommended Production Tools Stack

Author: Engineering Candidate • Project: AI Study Companion • Standard Technical Documentation

Part A: As Stated in the PRD | Part B: Recommended Real-World Tools Stack

Part A — AI Tools As Stated in the PRD

A1. Development-time AI (Used to Build the Product)

(PRD Section 20.5)

- **Coding Assistants:** GitHub Copilot / LLM coding assistants for rapid boilerplate generation, schema validation, and test scaffolding.
- **Debugging Tools:** Automated stack trace inspection, SSE token streaming diagnostics, and state desynchronization debugging.
- **Design Tools:** CSS Token formulation for dark/light themes, typography scales, glassmorphism visual hierarchy, and SVG vector illustrations.
- **Development Workflows:** Automated developer scripts for continuous test execution, static analysis, and multi-file refactoring.

A2. Runtime AI (Used by the Final Product)

(PRD Section 20.5)

- **Tutor:** Interactive conversational partner executing project-grounded RAG with strict citation generation (`Source: Doc` - `Page N`).
- **Quiz Generation:** Adaptive generation of MCQs and open-ended questions targeting weakest concepts and spaced-review schedules.
- **Assessment:** Dual-mode evaluation—deterministic matching for MCQs and 5-point rubric grading for qualitative open-ended responses.
- **Recommendations:** Context-aware study action generator answering "*What should I do next?*" anchored to specific document pages.
- **Document Understanding:** Background ingestion pipeline executing OCR, structural chunking, keyword extraction, and vector embedding generation.
- **Evaluation Models:** LLM-as-judge benchmarks quantifying groundedness, citation precision, and recommendation relevance.

A3. AI Operations to Abstract

(PRD Section 14 & Codebase `server/services/aiProvider.js`)

- **Text Generation:** Conversational completion with token-by-token streaming.
- **Structured Generation:** Strict JSON-schema validated outputs (quiz questions, rubric evaluations).
- **Embeddings / Retrieval:** Vector generation and cosine similarity matching scoped by `project_id` .
- **Evaluation:** Automated rubric grading and diagnostic test-case verification.
- **Document Understanding:** Multi-format document parsing (.pdf, .docx, .md, .txt) and concept graph extraction.

A4. Development Prompts — Required Organization

(PRD Section 20.6 & `PROMPTS.md`)

All development prompts are cataloged in `PROMPTS.md` categorized across the 8 disciplines:

1. Architecture
2. Frontend
3. Backend
4. Database

- 5. AI & RAG
- 6. Debugging
- 7. Testing
- 8. Documentation

A5. Key PRD Statement

"The exact models and providers are left to the candidate." — Section 14

"The exact architecture and technology stack are intentionally open." — Section 17

Note: The PRD does not mandate any single proprietary vendor. Model and tool selection is flexible, encouraging modular design.

Part B — Recommended AI Tools Stack (Gemini-Based Implementation)

B1. Core LLM (Tutor, Quiz, Grading, Recommendations)

Tool	Rationale & Capabilities
Gemini 1.5 / 2.0 Pro	1M–2M token context window holds large course materials, full conversation history, and learner profiles without aggressive compression; superior reasoning for grounded citations and nuanced grading.
Gemini Flash	High throughput, sub-200ms latency, and ultra-low cost; ideal for background concept tagging, classification, and summarization.

B2. Multimodal Document Understanding

Tool	Rationale & Capabilities
Gemini Native Multimodal	Natively understands diagrams, formulas, tables, and images directly from PDF pages without separate OCR fragmentation.
PyMuPDF / pdf-parse	Fast, lightweight text/metadata extraction feeding into structure-aware semantic chunking.
Unstructured.io (Fallback)	Clean layout-aware table and columnar extraction for complex academic papers.

B3. Embeddings & Retrieval

Tool	Rationale & Capabilities
Google text-embedding-004	High retrieval quality, dense semantic embeddings aligned with Gemini models.
pgvector / In-Memory Vector Store	Fast cosine-similarity vector queries strictly scoped by <code>project_id</code> without external service overhead.
Qdrant (Scale Alternative)	Dedicated vector database option for enterprise-scale multi-million document deployments.

B4. AI Observability & Tracing

Tool	Rationale & Capabilities
Langfuse / Built-in AI Trace Inspector	Complete request tracing (Trace ID, Model, Latency ms, Token in/out, Cost USD, Prompts, Completions).
Google Cloud Trace / Vertex AI Monitoring	Native cloud metrics for latency percentiles (p50, p95, p99) and token spend quotas.

B5. AI Evaluation

Tool	Rationale & Capabilities
Gemini (as LLM-Judge)	Automated 4-pillar benchmark evaluation for groundedness, citation accuracy, and rubric grading consistency.
Promptfoo	Automated CLI regression testing runner across prompt iterations and model updates.
Ragas	RAG-specific evaluation metrics (faithfulness, answer relevance, context recall).

B6. Background Job / Async AI Orchestration

Tool	Rationale & Capabilities
BackgroundQueue / Celery + Redis	Asynchronous material processing, non-blocking mastery recalculation, and idempotent retry loops.
WorkflowEngine / Inngest	Event-driven workflow engine with automatic exponential backoff (max 3 retries) and dead-letter safety.

B7. Structured Output Validation

Tool	Rationale & Capabilities
JSON Schema Validator / Pydantic	Validates structured AI payloads (quiz options, correct index, concept weights) before writing to the database.
Instructor	Schema enforcement and automated self-correction retries for AI structured completions.

B8. Security & Prompt-Injection Testing

Tool	Rationale & Capabilities
SecurityGuard / Rebuff	Pattern detection neutralizing prompt injections (<code>ignore instructions</code> , <code>reveal system prompt</code>), XML delimiter isolation.
Garak / SAST Security Linter	Automated vulnerability scanner verifying prompt boundary integrity and zero hardcoded secrets.

B9. Recommended Final Stack Summary

Component	Production Stack Choice	Current Workspace Implementation
Primary LLM	Gemini 1.5/2.0 Pro + Gemini Flash	<code>aiProvider.js</code> (Gemini API + Neural Engine fallback)
Document Understanding	Gemini Multimodal + PDF parser	<code>documentProcessor.js</code> (Multi-format + PDF text extractor)
Embeddings	Google text-embedding-004 + pgvector	<code>vectorStore.js</code> (Project-scoped cosine similarity)
Observability	Langfuse / Cloud Trace	<code>observabilityService.js</code> + <code>AiTraceModal.jsx</code>
Evaluation	Gemini-as-judge + Promptfoo	<code>evaluationSuite.js</code> (4-pillar benchmarks)
Queue	Celery + Redis	<code>backgroundQueue.js</code> (In-memory event-driven queue)
Output Validation	Pydantic / Instructor	<code>aiProvider.js</code> (<code>generateStructured</code> schema validator)
Security Testing	Garak + SAST Linter	<code>securityGuard.js</code> + <code>scripts/security_linter.js</code>

B10. Model Selection Rationale

Task	Selected Model	Technical Rationale
Tutor Conversation	Gemini Pro	Long context enables full document reference; complex multi-hop reasoning ensures factual answers.
Document Understanding	Gemini Pro (Multimodal)	Directly perceives diagrams, flowcharts, and math notations without OCR distortion.
Quiz Generation	Gemini Pro (Structured)	High reliability adhering to strict JSON schemas with 4 distinct options and explanations.
Open-Ended Grading	Gemini Pro	Nuanced semantic evaluation across 5 rubric dimensions (accuracy, reasoning, gaps, feedback).
Concept Tagging & Keywords	Gemini Flash	High-speed, cost-optimized token consumption for low-complexity metadata extraction.